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1 A fun visual proof of a prime number fact

Theorem. An odd prime number can be written as the sum of two square numbers if and only if it is one more than a multiple of 4. Also, this can only be done in one way except for reordering the two squares. For example, 37 is prime and one more than $4 * 9$ and it can be written as $6^2 + 1^2$ and 43 is prime and not one more than a multiple of 4 and cannot be written as the sum of 2 squares.

Proof. The proof that every prime number that is 3 more than a multiple of 4 is not the sum of two squares is the easy part. An even number squared is a multiple of 4, and an odd number can be written as $2k+1$, so its square is $4k^2 + 4k + 1$ by simple algebra which is clearly 1 more than a multiple of 4. Therefore any square number is 0 or 1 more than a multiple of 4, so the sum of 2 square numbers can never be 3 more than a multiple of 4.

Now if we have an odd prime, it is not the sum of two odd numbers squared or two even numbers squared because otherwise it would be an even number. So we want to investigate whether it is the sum of an even number squared and an odd number squared. The even number can be written as $2k$ and therefore we are looking at $p = x^2 + (2k)^2 = x^2 + 4k^2$ where p is prime. We now write $p = x^2 + 4yz$ and we want to solve for when $y = z$. What happens is that if y is not z , we can interchange the order of y and z to get a pair of solutions. Therefore if we could somehow show that there were an odd number of different solutions with x, y and z positive integers and p an odd prime 1 more than a multiple of 4 to $p = x^2 + 4yz$, we would know there has to be at least one with y and z the same so we would be done.

Now here comes the fun part:

We represent the x^2 part as an $x * x$ grid square. We represent the $4yz$ part as 4 $y * z$ grid rectangles. We place them in a windmill configuration like figure 1.

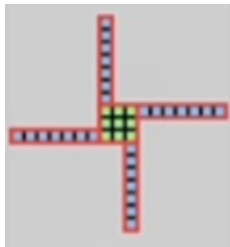


Figure 1

The convention used is such that figure 1 satisfies $x = 3, y = 1, z = 7$.

The area is clearly $p = x^2 + 4yz$ squares.

Lets suppose that y is the length of the side that touches the square, so in the image above $y = 1$. In cases like these where $y < \frac{x}{2}$, we can generate another solution by modifying the windmill diagram slightly as shown in figure 2:

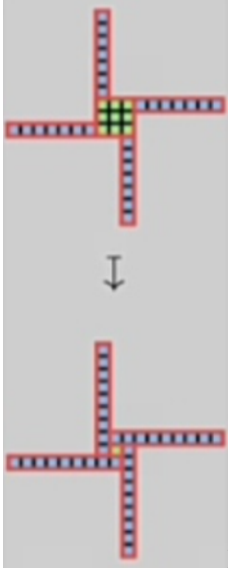


Figure 2

This transforms a solution of the $y < \frac{x}{2}$ case into the $y > z + x$ case. So we have a pairing between two solutions in the two cases.

Now lets look into the $\frac{x}{2} < y < x$ case and the $x < y < x + z$ case. They are also a pair, as shown in figure 3



Figure 3

If we are not in one of the exceptional $y = \frac{x}{2}$, $y = x$, $y = x + z$ cases then the following transformation rule corresponds to the pictures.

$$\begin{aligned} (x, y, z) &\mapsto (x - 2y, x + z - y, y) : y < \frac{x}{2} \\ (x, y, z) &\mapsto (2y - x, y, x + z - y) : \frac{x}{2} < y < x + z \\ (x, y, z) &\mapsto (x + 2z, z, y - x - z) : y > x + z \end{aligned}$$

Easy check is each of these transformations preserves $x^2 = 4yz$, gives a different solution, and gets you back to the same solution if you find which class the new value of y is in and apply this transformation again

So this leaves three cases. $y = \frac{x}{2}$, $y = x$, $y = x + z$. At this point you might realize that we have not yet used the fact that p is prime, which we need to do, otherwise you ought to be suspicious: If p is not prime, the theorem is not generally true (consider 21).

If $y = \frac{x}{2}$ then $x^2 + 4yz = x^2 + 2xz = x(x + 2z) = 2y(x + 2z)$ which is a problem as that implies $x^2 + 4yz$ is even contrary to p being odd.

If $y = x + z$ then our diagram will look like a square, and a square number is not prime.

If $y = x$, $x^2 + 4yz = x^2 + 4xz$, which is divisible by x . Therefore the only possibility is that $x = y = 1$. In this case, $z = \frac{p-1}{4}$ always gives us a solution that looks like the straight cross in figure 4.



Figure 4

To put it another way, p is 1 more than a multiple of 4, so write p as $1 + 4k$, so the straight cross with a block in the middle and four rectangles of length k will always exist. This is the only way that we do not have a pairing with another windmill diagram, as we have ruled out all the other cases.

Therefore, since there are a bunch of pairs and the straight cross, we have now shown that $p = x^2 + 4yz$ has an odd number of solutions. This means p can be written as the sum of two squares by the logic above.

Note that the solution corresponding to $y = z$ is, in general *different* from the straight cross solution. We count the solutions in two completely different ways.

□

The next theorem shows that the representation is unique, however the proof is nowhere near as interesting

Theorem. If a prime number is 1 more than a multiple of 4 then it can only be represented as the sum of two squares in one unique way except for reordering of the prime factors.

Proof. Side note: This can be done really elegantly by factoring $p = x^2 + y^2 = (x + iy)(x - iy)$ and using the fact that the complex integers factor uniquely (See the Groups, Rings and Modules course), but we do not assume that you know what this means or how to prove it, so I will now do this in a more elementary way.

So suppose $p = a^2 + b^2$ and $p = c^2 + d^2$ and $a \neq c, d, b \neq c, d$. Then one can check by expanding everything that

$$p^2 = (a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 \quad (1)$$

Also $p - a^2 = b^2$ and $p - c^2 = d^2$. Therefore $(p - a^2)d^2 = b^2d^2 = (p - c^2)b^2$. Therefore we can rearrange $(p - a^2)d^2 = (p - c^2)b^2$ to get

$$p(d^2 - b^2) = (ad)^2 - (bc)^2 = (ad - bc)(ad + bc)$$

One can check by expanding everything and moving all terms to one side that these equations are indeed equivalent. p is prime so it either divides $ad - bc$ or $ad + bc$. p cannot divide $ad - bc$ because by (1) if p did divide $ad - bc$, then if $ad - bc \neq 0$ we would know that $(ad - bc)^2 \geq p^2$ so $(ac + bd)^2 \leq 0$. This would force $ad + bc = 0$. Therefore if p did divide $ad - bc$ then either $ad - bc$ or $ad + bc$ would be 0 so we conclude that

$$p(d^2 - b^2) = (ad - bc)(ad + bc) = 0$$

Therefore in this case $d^2 = b^2$, hence contradicting the assumption that we have a non-unique solution. Otherwise, if p divides $ad + bc$, then first note that (1) by algebra is equivalent to

$$p^2 = (ad + bc)^2 + (ac - bd)^2$$

Therefore either $ad + bc = 0$, or we are in the situation where $p = ad + bc$ and $ac = bd$ as if both of these were false $(ad + bc)^2 + (ac - bd)^2$ would be greater than p^2 for the same reason as in the first case. If $ad + bc = 0$ then again

$$p(d^2 - b^2) = (ad - bc)(ad + bc) = 0$$

Again we conclude $d^2 = b^2$.

So the last case is when $ac = bd$. Note that the highest common factor of a and b must be 1, otherwise $p = a^2 + b^2$ is divisible by that factor squared, same for c and d. Therefore a and b do not share any prime factors and neither do c and d. Therefore in the equation $ac = bd$ all the prime factors in a have to go into d and all prime factors of d have to go into a, i.e. $a|d$ and $d|a$ so $a = d$, so there is no case where $a \neq c, d, b \neq c, d$.

□

2 Symmetric polynomials, roots of polynomials and algebraic numbers

Before we state the main theorem of this chapter, we need some definitions.

Suppose we have n variables $x_1, x_2, x_3, \dots, x_n$. Then s_k is defined as the sum of all products of k different variables. For example, $s_3 = x_1x_2x_3 + x_1x_2x_4 + \dots + x_{n-2}x_{n-1}x_n$.

These are called **elementary symmetric polynomials**. Note that s_k is exactly the coefficient of x^{n-k} in the polynomial $(x + x_1)(x + x_2)(x + x_3) \dots (x + x_n)$. So if $x_1, x_2, x_3, \dots, x_n$ were the actual roots, we would have the known result that the s 's are the coefficients up to possibly a change in the sign. We will use this result in the application after the proof.

Definition. A **symmetric polynomial** is a function of more than one variable such that the function is obtained by only adding and multiplying the variables or multiplying them by constants, and it has the property that if you input them in a different order you get the same function.

For example, $f(x, y, z) = x^2yz + xy^2z + xyz^2$ is a symmetric polynomial. This is because it is sums of products of x, y and z and $f(x, y, z) = f(x, z, y) = f(y, x, z) = f(y, z, x) = f(z, x, y) = f(z, y, x)$. However, something like $x^2 + y^2 - z^2$ is not symmetric as if you swap y and z you do not have the same function.

Note that our example $x^2yz + xy^2z + xyz^2 = s_1s_3$. This is a special case of the theorem.

Lemma. Suppose we have finite ordered lists of non-negative integers of a fixed length. Define an ordering on these lists where we say one list is greater than another if its first element is greater. If the first element is the same, say one list is greater if its second element is greater, and so on. Then the lemma claims that if I start with such a list, there is no infinitely decreasing chain of such lists.

Proof. Lets go by induction. We know for sure that for lists of 1 number this is true. For any number I start with I clearly cannot make an infinitely decreasing chain of non-negative integers starting from that number. So it is true for lists of length 1.

Now suppose it is true for k numbers. Then for a (k+1)-list, what happens is that I must decrease the first number after finitely many steps, because if I try to decrease the list, then by the induction hypothesis, after finitely many steps I will not be able to decrease the elements from positions 2 to k+1 anymore. At that point I will have to decrease the first element. But then this is a finite thing that happens finitely many times. So done.

□

Theorem. (the fundamental theorem of symmetric polynomials) All symmetric polynomials can be written as sums of products of elementary symmetric polynomials. Furthermore, if the coefficients of the symmetric polynomial are [rational/real/complex] then the coefficients of the expansion will also be [rational/real/complex], and this is evident from the proof.

Proof. Lets take each term in a symmetric polynomial like $Cx_1^{d_1}x_2^{d_2} \dots x_n^{d_n}$ and define the multidegree of that term as the list $(d_1, d_2, \dots, d_n)$, where some exponents may be 0. Sort the terms from largest to smallest multidegree where the ordering of the multidegree is as in the statement of the lemma.

Note that the list of the multidegree of the term with largest multidegree will be in non-increasing order. This is because if it was not, we could sort it into decreasing order and by symmetry of the polynomial that term must be one of the terms, and by how the ordering works it would have a larger multidegree so any unsorted list could not be the largest multidegree.

Note that the term with the largest multidegree in the product of symmetric polynomials is the product of the terms with the largest multidegree in the symmetric polynomials we are multiplying. This is because by how the ordering on the multidegree works, and the fact multiplying by any term just adds constants to the multidegree, multiplying by any term will not affect the ordering. If we let Y be the multidegree of the product of the terms in the polynomials we are multiplying that have largest multidegree, then all other terms in the polynomials will have lower multidegree so their product will have lower multidegree than Y – never higher – so Y is the largest.

Now consider

$$s_n^{d_n} s_{n-1}^{d_{n-1}-d_n} s_{n-2}^{d_{n-2}-d_{n-1}} \dots s_2^{d_2-d_3} s_1^{d_1-d_2}$$

Then the highest multidegree term in each of the terms we are multiplying will be $(d_n, d_n, \dots, d_n), (d_{n-1} - d_n, d_{n-1} - d_n, \dots, d_{n-1} -$

$(d_2 - d_3, d_2 - d_3, 0, \dots, 0), (d_1 - d_2, 0, \dots, 0)$, so the term with highest multidegree in the product will have multidegree $(d_1, d_2, \dots, d_n)$.

Therefore let $P(x_1, x_2, \dots, x_n)$ be a symmetric polynomial whose term with largest multidegree is $Cx_1^{d_1}x_2^{d_2}\dots x_n^{d_n}$. Then $P(x_1, x_2, \dots, x_n) - Cs_n^{d_n} s_{n-1}^{d_{n-1}-d_n} s_{n-2}^{d_{n-2}-d_{n-1}} \dots s_2^{d_2-d_3} s_1^{d_1-d_2}$ is a symmetric polynomial whose largest multidegree is at most $(d_1, d_2, \dots, d_n)$. But the largest multidegree is smaller as by construction the terms with that multidegree cancelled. Lets call this new polynomial $Q(x_1, x_2, \dots, x_n)$ and do the same procedure. But each time we do this, the largest multidegree decreases, so after finitely many steps it will be down to 0 by the lemma. So eventually we will have

$$P(x_1, x_2, \dots, x_n) - Cs_n^{d_n} s_{n-1}^{d_{n-1}-d_n} s_{n-2}^{d_{n-2}-d_{n-1}} \dots s_2^{d_2-d_3} s_1^{d_1-d_2} - (\text{More sums and products of elementary symmetric polynomials}) = 0$$

So move everything but P to 1 side then we are done.

Note that the remainder eventually becomes constant or 0, and when this happens we are done, although 0 has no largest multidegree.

□

Suppose we have complex numbers x, y, z such that the following is true:

$$x + y + z = 1$$

$$x^2 + y^2 + z^2 = 2$$

$$x^3 + y^3 + z^3 = 3$$

And we want $x^4 + y^4 + z^4$.

Then lets write $x^2 + y^2 + z^2$ in terms of the s 's. Write it as $x^2y^0z^0 + x^0y^2z^0 + x^0y^0z^2$. The largest multidegree is $(2,0,0)$ so by the proof we want to try subtracting s_1^2 .

$$x^2 + y^2 + z^2 - s_1^2 = -2xy - 2yz - 2xz$$

Now we have reduced the largest multidegree to $(1,1,0)$. In this case we want to add $2s_2$ to both sides, then we will get $x^2 + y^2 + z^2 - s_1^2 + 2s_2 = 0$ so

$$x^2 + y^2 + z^2 = s_1^2 - 2s_2$$

But $s_1 = 1$ from the first equation and $s_1^2 - 2s_2 = 2$ from the second one. Therefore $s_2 = -\frac{1}{2}$.

We can do the same procedure and it turns out that

$$x^3 + y^3 + z^3 - s_1^3 = -3x^2y - 3x^2z - 3y^2x - 3y^2z - 3z^2x - 3z^2y - 6xyz$$

$$x^3 + y^3 + z^3 - s_1^3 + 3s_1s_2 = 3xyz$$

$$x^3 + y^3 + z^3 = s_1^3 - 3s_1s_2 + 3s_3$$

So $s_3 = \frac{1}{6}$.

We could try to express $x^4 + y^4 + z^4$ in terms of elementary symmetric polynomials the same way and then solve for this. If we do this we get

$$x^4 + y^4 + z^4 = s_1^4 - 4s_1^2s_2 + 2s_2^2 + 4s_1s_3 = \frac{25}{6}$$

Application to algebraic numbers and roots of polynomials

Suppose we want to find a polynomial with integer coefficients such that $2^{\frac{1}{2}} + 3^{\frac{1}{3}}$ is a root, or more generally the sum of the roots of any two other polynomials is a root. The above are roots of $x^2 - 2$ and $x^3 - 3$ respectively. Let the roots of $x^2 - 2$ be x_1, x_2 and the roots of $x^3 - 3$ be y_1, y_2, y_3 , then there is a polynomial whose roots are all the possible sums of roots of the two polynomials, i.e. a polynomial whose roots are $x_1 + y_1, x_2 + y_2, x_1 + y_2, x_2 + y_1, x_1 + y_3, x_2 + y_3$. Let's call these $z_1, z_2, \dots, z_6$. Then notice that $(t - z_1)(t - z_2) \dots (t - z_6)$'s coefficients are exactly the elementary symmetric polynomials in the z variables. Permuting the x_i only reorders the pairwise sums, and so does permuting the y_j . Thus, every coefficient is separately symmetric in the two groups. Apply the elementary-symmetric-polynomial argument first to the x_i , treating polynomials in the y_j as coefficients. The leading-term subtraction preserves symmetry in the y_j . Apply the argument again to those coefficient polynomials. The coefficients will then be symmetric polynomials in the x and y variables in a way that can be expressed in terms of the elementary ones, which are integers, because they are (possibly with a sign change) the coefficients of the starting polynomials, i.e. $t^2 - 2$ and $t^3 - 3$. So $(t - z_1)(t - z_2) \dots (t - z_6)$ has integer coefficients and has $2^{\frac{1}{2}} + 3^{\frac{1}{3}}$ as a root. $t^6 - 6t^4 - 6t^3 + 12t^2 - 36t + 1$ is the expanded polynomial, by the way, which indeed has $2^{\frac{1}{2}} + 3^{\frac{1}{3}}$ as one of its roots. We could find this but it would take a lot of annoying work. One way to do this is to do $\left((t - \sqrt{2})^3 - 3\right) \left((t + \sqrt{2})^3 - 3\right)$. The same applies to products of two roots.

Now, note that if a polynomial has integer coefficients, we can divide through by the leading one. Therefore, algebraic numbers are exactly those that are roots of polynomials with rational coefficients and leading coefficient 1.

Now you know that algebraic numbers are closed under addition and (by a similar argument) multiplication and thus they are a group under addition and (if you remove zero) a group under multiplication as well (they have multiplicative inverses by reversing the order of the coefficients - do you see why? and additive inverses by substituting x with $-x$).

3 Two analysis constructions

3.1 A continuous function that is nowhere differentiable

I would try to motivate the construction, but I kind of want you to try to imagine what the graph might look like in your head of the function I show below so you see what the idea is. The 13 in the function in the theorem statement below looks random but it is just there to make the proof easier - you will see.

Theorem. The function $\sum_{n=0}^{\infty} \frac{\cos(13^n \pi x)}{2^n}$ is continuous everywhere

Proof. It is easy to see that if we replace ∞ in the sum then we will have a continuous function.

It will be easy to see that we have a continuous function if the convergence is uniform. We can establish this by making the tail small in a way that does not depend on x .

Note that

$$\left| \sum_{n=k+1}^{\infty} \frac{\cos(13^n \pi x)}{2^n} \right| \leq \sum_{n=k+1}^{\infty} \left| \frac{\cos(13^n \pi x)}{2^n} \right| \leq \sum_{n=k+1}^{\infty} \left| \frac{1}{2^n} \right| = \frac{1}{2^k} \rightarrow 0$$

We conclude as the uniform limit of continuous functions is continuous.

□

We claim that this function is nowhere differentiable. We will call this function $W(x)$. This function is “the” Weierstrass function, although it is a family of functions where we can change “13” and “2” around subject to some conditions.

If you have not seen this before, try to imagine what a nowhere differentiable function might look like. It is certainly difficult, since any attempt you make at drawing a continuous line will give you a line that has a slope. After the proof I will show an image of the graph.

To do this, for any x we will find a sequence of increments h_m such that

$$\left| \frac{W(x + h_m) - W(x)}{h_m} \right| \rightarrow \infty \text{ as } m \rightarrow \infty$$

We will define

$$S_m(x) := \sum_{n=0}^{m-1} \frac{\cos(13^n \pi x)}{2^n}$$

$$R_m(x) := \sum_{n=m}^{\infty} \frac{\cos(13^n \pi x)}{2^n}$$

So we have $S_m(x) + R_m(x) = W(x)$ for all m, x .

Now we will start the proof.

Theorem. $W(x)$ is nowhere differentiable.

Proof. Note that we cannot just differentiate each term in the sum since, as we showed in Level 4, this does not always work for infinite sums.

For a fixed x and an integer $m > 0$ write $13^m x = \alpha_m + e_m$ where α_m is the closest integer to $13^m x$ (pick the smaller one if there is a tie) so that $-\frac{1}{2} < e_m \leq \frac{1}{2}$

We will define

$$h_m := \frac{1 - e_m}{13^m}$$

This clearly $\rightarrow 0$ since the numerator is bounded but the denominator $\rightarrow \infty$. We will later use a bound that $\frac{1}{|h_m|} > \frac{2 \cdot 13^m}{3}$ which follows from the fact that the numerator is upper bounded by $\frac{3}{2}$.

We know that $S_m(x)$ is differentiable with derivative

$$S'_m(x) = - \sum_{n=0}^{m-1} \pi \left(\frac{13}{2} \right)^n \sin(13^n \pi x)$$

We have

$$|S'_m(x)| \leq \sum_{n=0}^{m-1} \pi \left(\frac{13}{2} \right)^n \leq 2 \frac{\left(\frac{13}{2} \right)^m}{11} \pi$$

The derivative bound holds at every point. By the mean value theorem, it also bounds the absolute value of every secant slope.

$$\left| \frac{S_m(x + h_m) - S_m(x)}{h_m} \right| \leq 2 \frac{\left(\frac{13}{2} \right)^m}{11} \pi$$

It is not clear why we need this but we will see later.

Now we note that if $n \geq m$ then

$$13^n(x + h_m) = 13^{n-m} 13^m \left(x + \frac{1 - e_m}{13^m} \right) = 13^{n-m} (13^m x + 1 - e_m)$$

$$= 13^{n-m} (\alpha_m + 1)$$

Since 13^{n-m} is odd, $13^n(x + h_m)$ is an integer equal to $(1 + \alpha_m) \bmod 2$. This is very good because now we know that

$$\cos(13^n \pi (x + h_m)) = \cos(\pi (\alpha_m + 1)) = -(-1)^{\alpha_m}$$

Now consider $R_m(x + h_m)$. This equals

$$\sum_{n=m}^{\infty} \frac{\cos(13^n \pi (x + h_m))}{2^n} = \sum_{n=m}^{\infty} \frac{-(-1)^{\alpha_m}}{2^n} = -(-1)^{\alpha_m} 2^{1-m}$$

Next we evaluate this tail at the original x . We have

$$\begin{aligned} R_m(x) &= \frac{\cos(13^m \pi x)}{2^m} + \sum_{n=m+1}^{\infty} \frac{\cos(13^n \pi x)}{2^n} \\ &= \frac{\cos(e_m \pi + \alpha_m \pi)}{2^m} + \sum_{n=m+1}^{\infty} \frac{\cos(13^n \pi x)}{2^n} \\ &= \frac{\cos(e_m \pi)}{2^m} (-1)^{\alpha_m} + \sum_{n=m+1}^{\infty} \frac{\cos(13^n \pi x)}{2^n} \end{aligned}$$

Now we have that

$$\begin{aligned} R_m(x + h_m) - R_m(x) &= -(-1)^{\alpha_m} 2^{1-m} - \frac{\cos(e_m \pi)}{2^m} (-1)^{\alpha_m} - \sum_{n=m+1}^{\infty} \frac{\cos(13^n \pi x)}{2^n} \\ &= -(-1)^{\alpha_m} \left(2^{1-m} + \frac{\cos(e_m \pi)}{2^m} + (-1)^{\alpha_m} \sum_{n=m+1}^{\infty} \frac{\cos(13^n \pi x)}{2^n} \right) \end{aligned}$$

Let A be the stuff inside the brackets. Then we know that

$$A \geq 2^{1-m} + \frac{\cos(e_m \pi)}{2^m} + \sum_{n=m+1}^{\infty} \frac{-1}{2^n} = 2^{-m} + \frac{\cos(e_m \pi)}{2^m} = 2^{-m} (1 + \cos(e_m \pi))$$

Since e_m is between $-\frac{1}{2}$ and $\frac{1}{2}$ we know that $\cos(e_m \pi) \geq 0$ so $A \geq \frac{1}{2^m}$.

This means that

$$|R_m(x + h_m) - R_m(x)| \geq \frac{1}{2^m}$$

Thus

$$\left| \frac{R_m(x + h_m) - R_m(x)}{h_m} \right| \geq \frac{1}{2^m |h_m|} \geq \frac{2}{3} \left(\frac{13}{2} \right)^m$$

By our bound on $\frac{1}{|h_m|}$ from earlier.

We know that

$$\frac{R_m(x + h_m) - R_m(x)}{h_m} = \frac{W(x + h_m) - W(x)}{h_m} - \frac{S_m(x + h_m) - S_m(x)}{h_m}$$

So the triangle inequality gives

$$\left| \frac{R_m(x + h_m) - R_m(x)}{h_m} \right| \leq \left| \frac{W(x + h_m) - W(x)}{h_m} \right| + \left| \frac{S_m(x + h_m) - S_m(x)}{h_m} \right|$$

Thus

$$\left| \frac{W(x + h_m) - W(x)}{h_m} \right| \geq \left| \frac{R_m(x + h_m) - R_m(x)}{h_m} \right| - \left| \frac{S_m(x + h_m) - S_m(x)}{h_m} \right|$$

By our bounds from earlier we get

$$\begin{aligned} \left| \frac{W(x + h_m) - W(x)}{h_m} \right| &\geq \frac{2}{3} \left(\frac{13}{2} \right)^m - 2 \frac{\left(\frac{13}{2} \right)^m}{11} \pi \\ &= \left(\frac{13}{2} \right)^m \left(\frac{2}{3} - \frac{2}{11} \pi \right) \end{aligned}$$

Since $\pi < \frac{11}{3}$ this is not negative (There are many ways to prove this, including by considering the perimeter of a regular hexagon outside a circle being $4\sqrt{3}$ which is between 2π and $\frac{22}{3}$ geometrically and by squaring $4\sqrt{3}$ and $\frac{22}{3}$ and comparing them), we conclude that the difference quotient does indeed go to infinity for this sequence h_m , so W is continuous but nowhere differentiable.

□

Now we show images.

Figure 5 shows the graph of $W(x)$ ($-\frac{\pi}{2} < x < \frac{3\pi}{2}$).

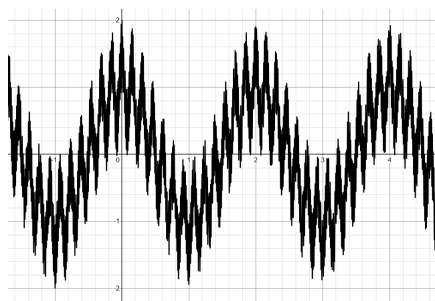


Figure 5

Figure 6 shows the graph zoomed in a lot ($0.45409 < x < 0.45413$)

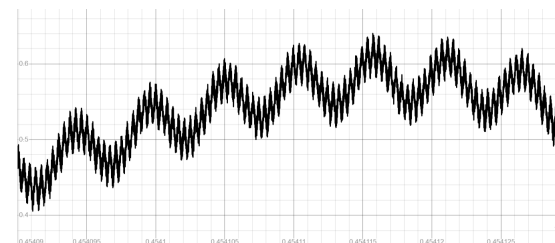


Figure 6

Interesting to see how what we proved is reflected in the pictures! The graph essentially is a fractal curve that looks the same at all magnifications.

3.2 A differentiable function with a bounded non-Riemann-integrable derivative

We will aim to construct a function on $[0, 1]$ with the following properties:

- It is differentiable everywhere (with either one-sided derivatives at the endpoints or with extension to an open interval around $[0, 1]$)
- Its derivative is bounded
- Its derivative is not Riemann integrable

To do this we will invoke a theorem from Analysis which says that a bounded function is Riemann integrable if and only if it is discontinuous on a set of measure 0. Therefore, we aim to make a function which has a derivative that is bounded but discontinuous on a set which is not measure 0.

First, it is not even easy to come up with a derivative with a single point of discontinuity. If you try to make a jump discontinuity, what will happen is your function just won't even be differentiable at that point. For a function to genuinely be differentiable yet its derivative is not continuous at said point, is not trivial. But my example of how we can make a single point discontinuity will be useful for better understanding the problem.

Our example of a derivative with one point of discontinuity will be the derivative of the function $f(x) = x^2 \sin\left(\frac{1}{x}\right)$ and $f(0) = 0$. Figure 7 shows the graph of what this looks like:

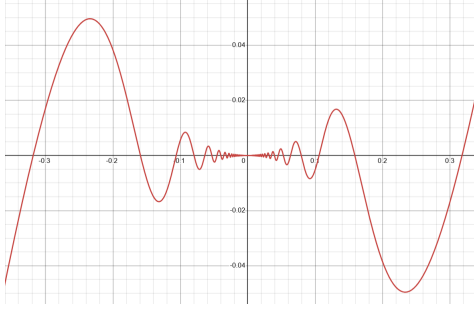


Figure 7

This is clearly differentiable away from 0 with derivative $2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$ by the product and chain rules.

At 0, we note that as $h \rightarrow 0$,

$$\left| \frac{f(h) - f(0)}{h} \right| = \left| \frac{f(h)}{h} \right| = \left| h \sin\left(\frac{1}{h}\right) \right| \leq |h| \rightarrow 0$$

Thus we have $f'(0) = 0$. Note though that near 0, $2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$ will behave wildly, as what happens is $2x \sin\left(\frac{1}{x}\right)$ approaches 0 but $\cos\left(\frac{1}{x}\right)$ bounces between -1 and 1 faster and faster as $\frac{1}{x} \rightarrow \infty$. Figure 8 shows a graph of the derivative and it is easy to see that there is no way that the derivative will be continuous at 0, despite existing everywhere. The trick was essentially to make the derivative oscillate like this since a simple jump discontinuity will make the function not be differentiable in the first place.

In fact, it is possible to show that a derivative has the intermediate value property i.e. that if f is differentiable on an $[a, b]$ and $f'(a) < c < f'(b)$ where $a < b$ then there exists $a < x < b$ with $f'(x) = c$. The idea of how to prove this is by noting that f is continuous, so as end points between a and b vary continuously so will the secant slope, which must attain c by the intermediate value theorem and thus by the mean value theorem the derivative attains c if we start from the secant slopes of $[a, a]$ and $[b, b]$ which we will take to be $f'(a)$ and $f'(b)$ and making sure the endpoints apart at all points in between, and continuity of the secant slope is retained at the end points by definition of the derivative.

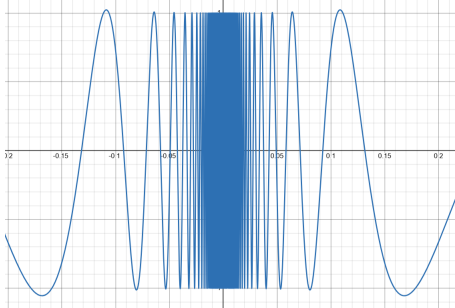


Figure 8

Note that if we drop the “bounded” condition, the construction becomes much easier: A function like $f(x) = x^2 \sin\left(\frac{1}{x^2}\right)$ will be differentiable for the same reason but if we compute its derivative we see that it will be unbounded, and unbounded functions are not Riemann integrable as we define Riemann integrable functions to have to be bounded, however the improper integral of $f'(x)$ in this case still exists.

For convenience, we will use $g(x) = \begin{cases} x^2 \sin\left(\frac{\pi}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$ as a basis for our construction. This satisfies $g(1) = 0$. We will take the piece of g from 0 to 1 and rotate it by 180 degrees about the point $(1, 0)$ to get a function on $[0, 2]$ that is differentiable everywhere but its derivative oscillates wildly near 0 and 2. Figure 9 shows this function, which we will use for our construction.

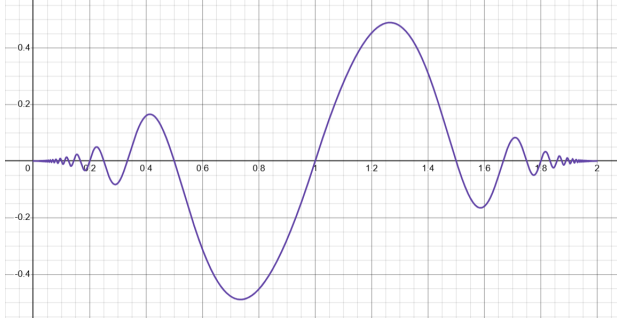


Figure 9

This is differentiable at the point of rotation with the original derivative, and this is clear. Note that $g\left(\frac{1}{k}\right)$ for any positive integer k is also 0 so we could rotate 180 degrees about these points as well and still have a valid similar thing on $[0, \frac{2}{k}]$. We will use this.

Before we do the construction I will construct the set that we aim to make the derivative discontinuous on and prove that this set does not have measure 0.

Figure 10 shows a construction of the Cantor set. The way this works is we start with $[0, 1]$ and then remove the middle third so we end up with 2 closed intervals $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$. We will remove the middle third from each of these intervals to get 4 closed intervals, and continue. We will then take the intersection of all these closed sets and we will get the Cantor set. The Cantor set will contain the endpoints, such as $\frac{1}{3}$ and $\frac{2}{9}$, as well as some other points like $\frac{1}{4}$ that did not get removed, as it is possible to show that it is in the left branch of the main set and the right branch of that and the left branch of that and the right branch of that and so on.

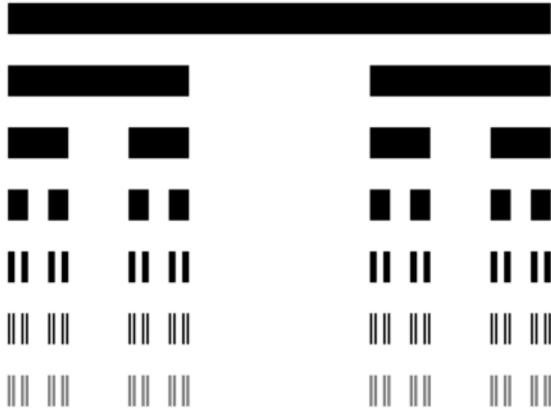


Figure 10

The issue, though, is that this set has measure 0. However, there is a nice modification we can make.

We can construct the “fat Cantor set” by removing intervals of size $\frac{1}{4^n}$ from the middle of each closed interval on the n ’th step, as opposed to $\frac{1}{3^n}$. See Figure 11 for a visual of the fat Cantor set.



Figure 11

We will make a function which has a derivative which is not continuous anywhere on the fat Cantor set, but yet the derivative is bounded. By the next proposition, it will follow that the derivative of the function we will construct, if we successfully construct it, will indeed be non-integrable.

Proposition. The fat Cantor set does not have measure 0.

Proof. We will prove that there is no set of open intervals that cover the fat Cantor set such that their total length is $< \frac{1}{2}$. Since it is not possible to have uncountably many open intervals with each non-zero length and finite total length (I've proven this somewhere in level 6 or 7), we can assume for a contradiction that there are countably many open intervals with this property.

Take a fictitious such covering of the fat Cantor set with open intervals. Now add to this covering all the open intervals we remove from the construction of the fat Cantor set to get a cover of the closed set $[0, 1]$ by open intervals. This will include the interval $(\frac{3}{8}, \frac{5}{8})$ as well as 2 intervals of length $\frac{1}{16}$, 4 of length $\frac{1}{64}$, and at the n 'th step it will include 2^{n-1} intervals each of length $\frac{1}{4^n}$. The total length of the intervals we will add to get a cover for $[0, 1]$ will therefore be $\sum_{r=1}^{\infty} \frac{2^{r-1}}{4^r} = \frac{1}{2} \sum_{r=1}^{\infty} \frac{1}{2^r} = \frac{1}{2}$.

Therefore, by our contradiction assumption, we end up with a covering of all of $[0, 1]$ by open intervals of total length $\frac{1}{2} + (< \frac{1}{2})$ which is < 1 .

Now you may think this is already a contradiction so we are done, and that would be true, but it is not obvious since this fails if we consider the set of rational numbers between 0 and 1 in which you may come to the conclusion that it cannot have measure 0 even though I could prove to you that in fact it does, so we have to be careful.

So we have this weird covering of $[0, 1]$ with total length < 1 . Since $[0, 1]$ is a closed and bounded subset of $\mathbb{R}$, it has the property that every open cover of $[0, 1]$ has a finite subcover. Thus, take a finite subcover of the cover above. This will be a FINITE cover of $[0, 1]$ with total length < 1 , which is not possible since overlaps can only reduce the length of the union and we no longer can have any infinite sum weirdness. Therefore we conclude.

□

Now we begin the construction.

First, we make our function be 0 everywhere on the fat Cantor set.

We essentially make a modification of figure 9 where instead of using 1 as our point to rotate about we use $\frac{1}{8}$. We then place this modification, which is of total length $\frac{1}{4}$, into the first interval we remove which is $(\frac{3}{8}, \frac{5}{8})$. We then do another modification where now the point we rotate about is $\frac{1}{32}$. This gives us something analogous to figure 9 which we will place into the 2 intervals we removed at the second step of the fat Cantor set construction. We will do the same thing; In general we will modify figure 9 such that the point we rotate about is $\frac{1}{2 \cdot 4^n}$ and place it in each of the 2^n intervals we removed at the n 'th step in the fat Cantor set construction.

The resulting function after we do this bizarre construction is Volterra's function. Figure 12 shows a sketch of this, although it is difficult to see the inserted things past the second step, since they get very small.

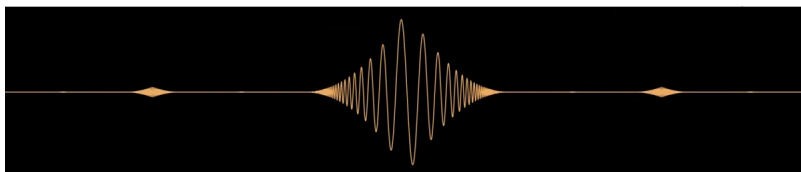


Figure 12

If we zoom in we will see more truncated and rotated pieces appear in the smaller gaps.

The function is clearly differentiable inside these open intervals i.e. not in the Cantor set. However, what is also clear is that near the 2 boundary points of each of these open intervals, the derivative will oscillate wildly.

Theorem. At any point in the fat Cantor set, Volterra's function has derivative 0, and yet the derivative is discontinuous. Furthermore, its derivative is bounded.

Proof. To see that it has derivative 0 at x for x in the fat Cantor set note that $\frac{f(x+h)-f(x)}{h} = \frac{f(x+h)}{h}$. If $x+h$ is in the fat Cantor set this is 0, otherwise it is in one of the open intervals. If $h > 0$ we will let k be such that $x+k$ is the left boundary of this open interval, and if $h < 0$ we will let k be such that $x+k$ is at the right boundary of this open interval.

Clearly, k lies between 0 and h so we have

$$\left| \frac{f(x+h)}{h} \right| \leq \left| \frac{f((x+k)+h-k)}{h-k} \right|$$

But as $h \rightarrow 0$, $h-k$ also goes to 0. Crucially, we know that by our construction

$$\left| \frac{f((x+k)+h-k)}{h-k} \right| \leq \frac{|h-k|^2}{|h-k|} \rightarrow 0$$

Thus the difference quotient goes to 0 so the derivative is 0. The last inequality is since any rotation of figure 9 will always be bounded above and below by the parabolas $\pm(z-(x+k))^2$ where z is the independent variable, and this is independent of which interval $x+h$ is in, hopefully this makes sense.

Now we just need to show that the derivative is discontinuous. For any x in the fat Cantor set, note that it is clear that after the n 'th step it is true that any point in the set from the n 'th iteration of the construction is within a distance of $\frac{1}{2^n}$ units from a removed interval. Therefore, in the final function, in any interval about x , no matter how small, it will contain an area where the derivative oscillates wildly (for example, consider the midpoint of removed intervals from the fat Cantor set, where the derivative is $-\pi$ by direct calculation and these points get arbitrary close to x), so the derivative will not be continuous there.

To show that the derivative is bounded, note that outside the fat Cantor set it follows from boundedness of the derivative of g on any finite interval containing 0 and inside the fat Cantor set it is 0.

□

The derivative is bounded and its discontinuity set is the fat Cantor set, which does not have measure zero. The Riemann-integrability criterion therefore shows that V' is not Riemann integrable.

This completes the construction and proof. If you're like me you'll think wow this is so cool and fun. If you think what's the point because we're only interested in well behaved stuff, then, well, I don't know what to tell you. These examples show why hypotheses for theorems matter and why I get annoyed when physicists pretend like they don't.